USER MANUAL · REV 1.0

BIG MUFF

Dual-Preset MIDI Fuzz · a faithful clone of the Big Muff-style fuzz (Green Russian era) · handmade in Austria by Walter Craffonara

NOTE This pedal is a faithful, circuit-for-circuit clone of the Big Muff-style fuzz of the Green Russian era, expanded with functions I added: relay-switched dual presets and MIDI. The name “Big Muff” is used only to identify the original circuit this build recreates. CRAFFONARA FX is an independent builder and is not affiliated with, authorised by or endorsed by Electro-Harmonix / New Sensor Corp.

1 Overview & Concept

At its core this is a faithful clone of the Big Muff-style fuzz of the Green Russian era — famous for its thick, saturated wall of fuzz. The clipping section can be switched between full fuzz and a tighter overdrive voicing.

What I added is the part the original never had: the Fuzz and Tone controls exist twice. A relay silently switches between the two pots of each channel — by soft-touch button on the pedal, or remotely via MIDI. Dial in a rhythm sound and a lead sound and recall both hands-free.

2 Controls & Connections

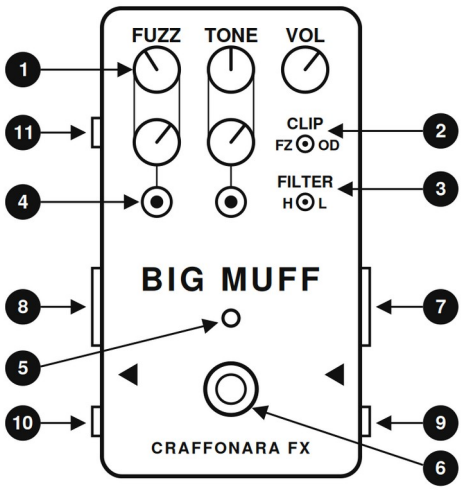


Fig. 1 — Panel and connections

1 FUZZ · TONE · VOL the main controls for fuzz amount, tone colour and output level. FUZZ and TONE each exist twice: an upper pot with a white marker and a lower pot with a red marker.

2 CLIP FZ = full Green-Russian fuzz, OD = a tighter, more focused overdrive voicing with less compression.

3 FILTER H = high peak for an aggressive voicing, L = low peak for a smoother, darker response.

4 PRESET buttons one soft-touch button per channel (Fuzz, Tone). LED ring off = preset #1 (upper pot), LED ring on = preset #2 (lower pot).

5 STATUS LED shows engaged / bypassed state and which preset is active.

6 FOOTSWITCH silent soft-touch switch, engages / bypasses the effect (relay true bypass).

7 INPUT 6.35 mm mono jack from your guitar or the previous pedal.

8 OUTPUT 6.35 mm mono jack to your amp or the next pedal.

9 MIDI IN TRS socket for incoming MIDI (Type A).

10 MIDI THRU TRS socket passing incoming MIDI on to the next device.

11 POWER 9 V DC, center-negative.

3 The Dual-Preset System

Each preset button toggles its channel between two independent pots. With the LED ring off, the upper pot with the white marker is active (preset #1); with the LED ring on, the lower pot with the red marker is active (preset #2).

Because Fuzz and Tone have their own buttons, the four combinations give you four distinct sounds from one pedal — for example a rhythm setting and a lead setting, recalled by foot or from a MIDI controller.

> **NOTE** Preset switching is relay-based and click-free. Only the selected pot sits in the signal path, so the unused pot causes no tone loss.

4 Voicing Controls

CLIP — FZ / OD FZ gives the full, compressed Green-Russian fuzz wall; OD tightens the clipping into a more focused overdrive with more note definition.

FILTER — H / L sets the tone-stack peak. H is aggressive and cuts through a dense mix; L is smoother and darker.

5 MIDI Reference

The pedal responds to MIDI Control Change (CC) messages on its MIDI channel (see the next section).

CC	Value	Command
00	00	TONE preset #1 and FUZZ preset #1
	01	TONE preset #2 and FUZZ preset #1
	02	TONE preset #1 and FUZZ preset #2
	03	TONE preset #2 and FUZZ preset #2
10	00	TONE preset #1
	01	TONE preset #2
	02	Toggle TONE preset #1 / #2
30	00	FUZZ preset #1
	01	FUZZ preset #2
	02	Toggle FUZZ preset #1 / #2

6 Setting the MIDI Channel

1. Disconnect the pedal from the power supply.
2. Press and hold the footswitch while connecting the power supply — the LED starts to flash.

3. Press the footswitch as many times as the channel number you want to set (e.g. three times for MIDI channel 3).
4. Press and hold the footswitch until the pedal switches off completely.
5. On the next power-up the pedal responds to the new MIDI channel.

7 Power

Run the pedal on 9 V DC, center-negative, from any regulated pedalboard supply. There is no battery compartment.

8 Specifications

Parameter	Value
Type	Fuzz — faithful clone of the Big Muff-style circuit (Green Russian era), with dual presets and MIDI
Controls	FUZZ ×2, TONE ×2 (relay-switched presets), VOL, CLIP toggle, FILTER toggle
Clipping	FZ = full fuzz, OD = tighter overdrive voicing
Presets	2 × Fuzz, 2 × Tone — relay-switched, four combinations
Bypass	Relay true bypass, silent soft-touch footswitch
MIDI	IN + THRU (TRS, Type A), Control Change
Audio I/O	6.35 mm mono input and output
Power	9 V DC, center-negative
Enclosure	Brushed aluminium, CNC-drilled (Hammond 1590B format)

NOTE Trademark notice: “Big Muff” and “Electro-Harmonix” are trademarks of New Sensor Corporation. They are used here purely to describe the original circuit that this independent build recreates. CRAFFONARA FX is not affiliated with, authorised by or endorsed by the trademark holder.